Practical Work No. 1 (PW1)

Data Models in Database Systems, 2025

SAMPLE OF PRACTICAL WORK REPORT

General information:

Group number:	DMDS PW group 99
Variant (domain):	HOSPITAL
Link to the video:	https://link-to-video.com/kjdkwheehdknwon
Link to the LiveSQL script:	https://livesql.oracle.com/ords/livesql/s/cj188tuz2xpgvlqk6srnxaquv

Group information:

#	Role	Full name	Student ID number
1	**Administrator**	Kaspars Gutmanis	888AAA222
2	**Member**	John Smith	999AAA111
3	**Member**	Marie Claire	000AAA888

1. PROJECT PASSPORT

1.1. TABLES

Criterion	Name of the table	LiveSQL Statement No.			Score
		CREATE	INSERT	SELECT	
Table with row type objects	INGREDIENTS	**2**	**3**	**4**	
	PEOPLE	**18**	**19**	**20, 21**	
	PRESCRIPTIONS	**23**	**24, 25**	**26**	
	MEDICATIONS	**12**	**13**	**14**	
Table with object columns	PEOPLE	**18**	**19**	**20, 21**	
	PRESCRIPTIONS	**23**	**24, 25**	**26**	
Table with object collections	MEDICATIONS	**12**	**13**	**14**	
Table of other type	-	-	-	-	

1.2. FUNCTIONS & OPERATORS

Function	LiveSQL Statement No.	Score
VALUE()	**14, 28, 31, 32**	
TABLE()	**14**	
REF()	**13, 24, 25, 27, 32, 35, 38**	

Function	LiveSQL Statement No.	Score
DEREF()	**14, 26, 30, 35, 38**	
TREAT()	**14, 21**	
IS OF type	**14**	

1.3. OBJECT METHODS

Criterion	Type name and function or procedure name	LiveSQL Statement No.			Score
		Declare	Define	Test	
Member function	MEDICATION get_description	**7**	**40**	**45**	
	INJECTION_MEDICATION get_description	**8**	**42**	**45**	
	TABLET get_description	**10**	**43**	**45**	
	LIQUID get_description	**11**	**44**	**45**	
	PATIENT GET_PRESCRIPTIONS	**15**	**38**	**39**	
	DOCTOR GET_SALARY	**16**	**35**	**37**	
	DOCTOR GET_PRESCRIPTIONS	**16**	**35**	**36**	
	PERSON GET_AGE	**17**	**32**	**33**	
	PERSON FETCH_PRESCRIPTIONS	**17**	**32**	**32**	
	PERSON GET_PRESCRIPTIONS	**17**	**32**	**34**	
Member procedure	PRESCRIPTION show	**22**	**27**	**28, 34**	
MAP member function	MEDICATION get_total_dosage	**7**	**40**	**41**	
ORDER member function	PRESCRIPTION compare	**22**	**27**	**31**	
Static function	-	-	-	-	
Static procedure	PRESCRIPTION ADD_PRESCRIPTION	**22**	**27**	**29, 30**	
Function returning SYS_REFCURSOR	DOCTOR GET_PRESCRIPTIONS	**16**	**35**	**36**	
	PERSON FETCH_PRESCRIPTIONS	**17**	**32**	**32**	

Criterion	Type name and function or procedure name	LiveSQL Statement No.			Score
		Declare	Define	Test	
	PERSON GET_PRESCRIPTIONS	**17**	**32**	**34**	

1.4. OTHER CONSTRUCTIONS

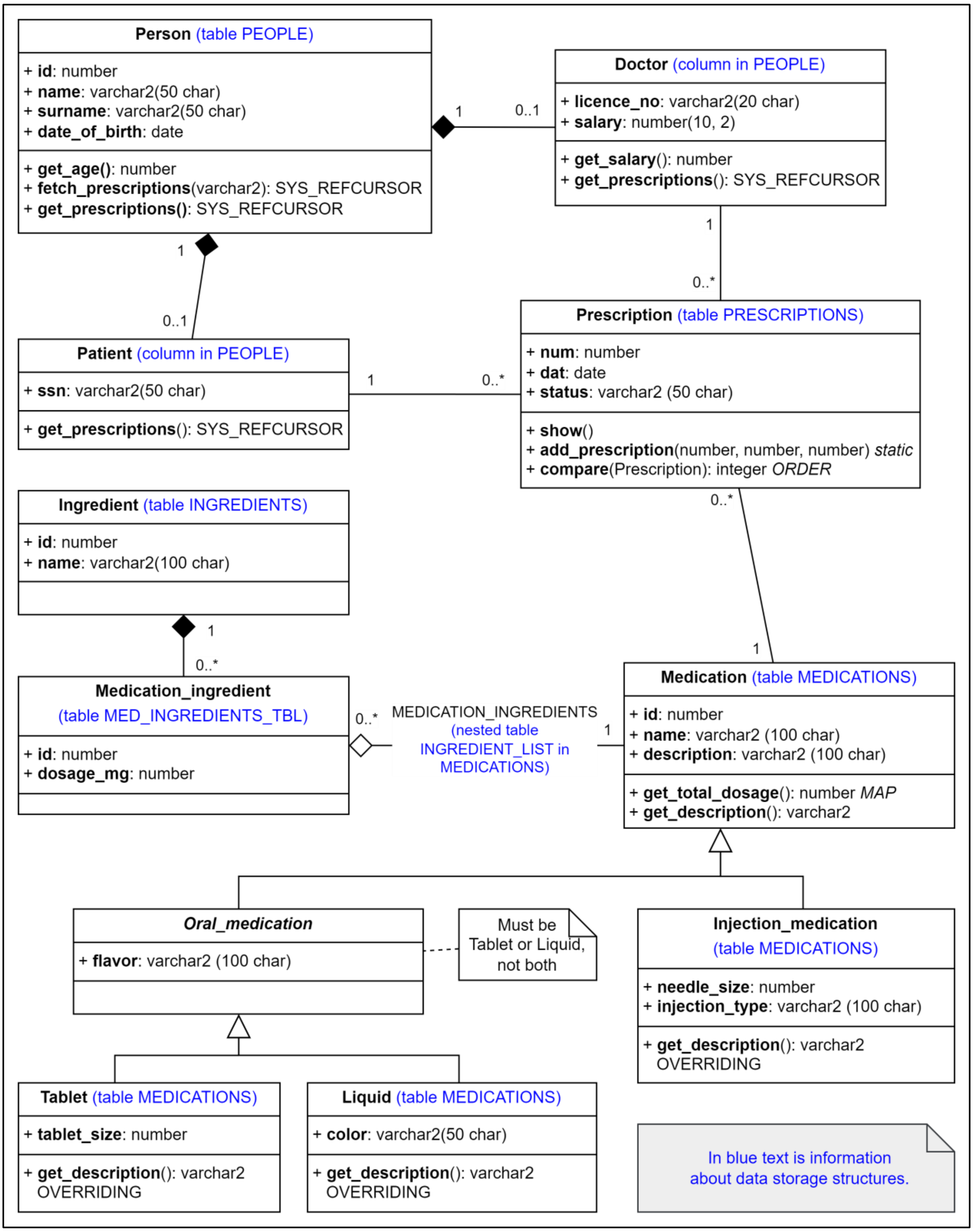
Type inheritance

- INJECTION_MEDICATION **UNDER** MEDICATION – statement No. 8
- ORAL_MEDICATION **UNDER** MEDICATION – statement No. 9
- TABLET **UNDER** ORAL_MEDICATION – statement No. 10
- LIQUID **UNDER** ORAL_MEDICATION – statement No. 11

Function or procedure overriding

- INJECTION_MEDICATION – get_description – statement No. 8
- TABLET – get_description – statement No. 10
- LIQUID – get_description – statement No. 11

2. CLASS DIAGRAM

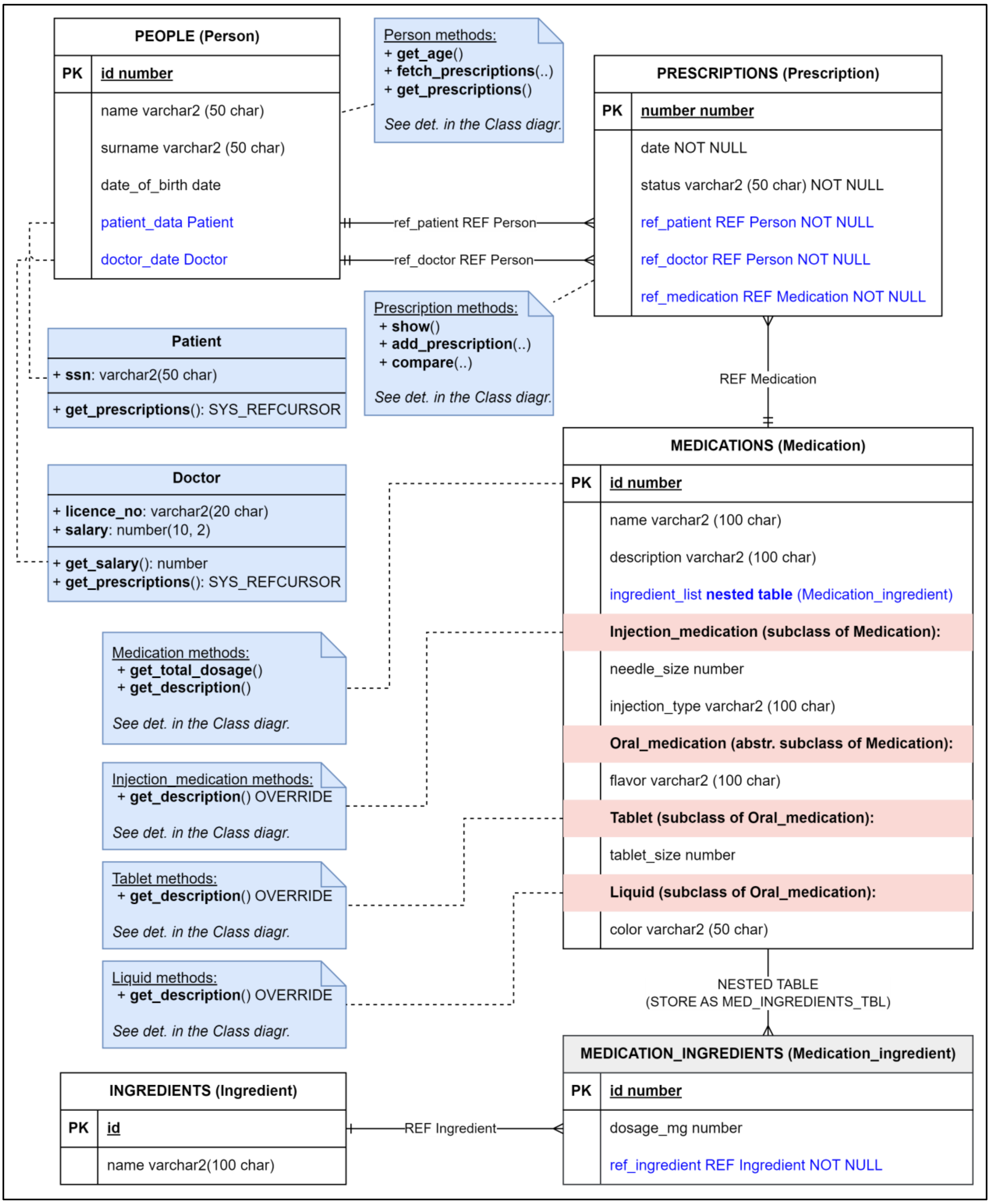


This class diagram models the structure of a hospital system database, focusing on the relationships between patients, doctors, medications, and prescriptions.

1. **Person**: This class represents general information about individuals in the hospital system. It serves as a base structure from which the classes *Patient* and *Doctor* draw shared attributes, though they are not formally subclasses.

2. **Patient** and **Doctor**: While not formal subclasses of *Person*, these classes represent specific types of individuals in the system. Each *Patient* and *Doctor* is connected to *Prescription*, illustrating that both patients and doctors can have multiple prescriptions associated with them. This setup supports a many-to-one relationship from *Prescription* to *Patient* and *Doctor*, as each prescription is linked to a specific patient and doctor, but both patients and doctors can be linked to many prescriptions over time.
3. **Prescription**: This class forms the core link between *Patient*, *Doctor*, and *Medication*. Each *Prescription* connects a single *Patient* and a single *Doctor* with specific medications prescribed for treatment. This design implies that each patient may have multiple prescriptions, each assigned to one doctor, with each prescription potentially including multiple medications.
4. **Medication**: This class captures the details of medications that can be included in prescriptions. It is linked to *Prescription* to indicate which medications are prescribed to a patient. The *Medication* class also connects to *Ingredient* through *Medication Ingredient*, allowing each medication to consist of various components.
5. **Medication Ingredient** and **Ingredient**: The *Medication Ingredient* class serves as a linking structure between *Medication* and *Ingredient*, supporting a many-to-many relationship. This structure allows each medication to include multiple ingredients, and each ingredient can be part of multiple medications, aligning with the real-world complexity of pharmaceutical composition.
6. **Oral Medication** and **Injection Medication**: These represent specific types of medications, demonstrating the flexibility to model different medication types. They extend *Medication* by adding specific attributes for different types, such as flavor for oral medications and needle_size for injections. This use of inheritance enables specialized properties for each medication type.

3. EXTENDED ER DIAGRAM



If you would like to provide explanations about the notation used in the diagram or any nuances that you think should be highlighted, it should be done here.

4. GROUP WORK REPORT

Here should be the group work report containing answers to the questions mentioned in the PW1 description.

CONCLUSIONS

Your conclusions should be provided here.

SOURCES AND TOOLS

In this section, you must list all sources (for internet sources, include the link) and technological tools used in the preparation of this work.

Sources:

1. …

Tools:

1. …